

Here is a beginner-friendly breakdown of all those technical terms, explained with real-world analogies and directly tied to how they work inside **SatQuery AI**.

1. Data Handling & Storage (The "Speed & Safety" Tools)

LMDB & Safetensors

- **Plain English Analogy:** Reading 500,000 separate image files from a standard computer folder is like opening and closing 500,000 physical photo albums one by one—it slows the computer to a crawl.
- **LMDB (Lightning Memory-Mapped Database):** A specialized database format that packs hundreds of thousands of satellite image pairs into one continuous file on disk 1. The computer reads data directly out of system RAM at lightning speed 1.
- **Safetensors:** A modern, secure file format for saving machine learning model weights (the "brain numbers" of an AI). Unlike older Python .pkl (pickle) files, .safetensors cannot accidentally run malicious code when loaded and loads into GPU memory almost instantly 1.
- **In SatQuery AI:** Used to store the **BigEarthNet.txt** dataset (~464,000 Sentinel-1/2 image pairs) so GPUs process batches without bottlenecks during fine-tuning 1-3.

ScoreRS Quality Filter

- **Plain English Analogy:** "Garbage in, garbage out." Public datasets collected automatically often include corrupted images, heavy cloud cover, or incorrect text labels.
- **How it works:** A small "quality-checker" AI scores every image-text pair from best to worst 1, 4. The bottom 25% (the noisiest or most blurry data) is discarded before training starts 1.
- **In SatQuery AI:** Filters out low-quality pairs from BigEarthNet.txt 1. Research shows that training an AI on 75% clean data yields a significantly smarter model than training on 100% noisy data 4, 5.

2. Image Preprocessing (Bridging the Satellite Gap)

GSD-Normalization Tiling

- **What is GSD? Ground Sample Distance (GSD)** is the real-world distance represented by a single pixel on screen 6.
- **Sentinel-2 (Training Data):** $1 \text{ pixel} = 10 \text{ to } 60 \text{ meters}$ on the ground 7, 8.
- **Cartosat-2S (ISRO Evaluation Data):** $1 \text{ pixel} = 0.65 \text{ meters}$ on the ground (much higher detail) 7, 8.
- **The Problem:** An AI trained to recognize a building that takes up 5×5 pixels in Sentinel imagery will fail when a building takes up 100×100 pixels in Cartosat imagery 8.
- **In SatQuery AI:** GSD-Normalization rescales, resamples, and tiles satellite images so that pixel sizes match the scale the model learned during training 8, 9.

Radiometric-Jitter Augmentation

- **Plain English Analogy:** Imagine taking a photo on an iPhone vs. a DSLR camera—colors, contrast, and digital noise look slightly different.
- **How it works:** Satellite sensors capture light and radar differently (optical colors vary, and SAR radar has "speckle noise"). During training, the pipeline intentionally alters ("jitters") the brightness, color balance, and radar noise of Sentinel images to simulate how Cartosat-2S optical and RISAT radar imagery look 9, 10.
- **In SatQuery AI:** Teaches the AI to remain reliable when tested on ISRO's Cartosat and RISAT satellites, preventing performance drops caused by sensor lighting differences 9, 10.

Content-Addressed Embedding Caching

- **Plain English Analogy:** A smart memory shortcut. Converting a heavy satellite image into numerical "AI visual tokens" takes significant GPU computing power.
- **How it works:** When an image is uploaded, the system generates a unique digital fingerprint (a hash) of the file's exact bytes 11, 12. If a user asks follow-up questions about the same image, the system reuses the cached "AI tokens" instantly instead of re-processing the image from scratch 11, 12.

3. Training Loss Functions (How the AI Learns from Mistakes)

A **Loss Function** is a mathematical penalty score measuring how incorrect an AI's guess was. Training aims to bring this score as close to zero as possible.

Autoregressive Token Cross-Entropy Loss ($\mathcal{L}_{\text{CE}}$)

- **How it works:** "Autoregressive" means generating output **one word (or token) at a time** 13. Cross-entropy checks the probability the AI assigned to the correct next word. If the right next word is "river" and the AI guessed "forest", it receives a high mathematical penalty 13.
- **In SatQuery AI:** Used to train the AI to generate natural-language answers for Visual Question Answering (VQA) and scene descriptions 13.

Sequence-to-Sequence Box Regression Loss ($\mathcal{L}_{\text{box}}$)

- **How it works:** Usually, object detectors use complex math boxes. In SatQuery AI, a bounding box is formatted as text coordinate tokens: [xLeft, yTop, xRight, yBottom] 13, 14. The AI learns to predict bounding box numbers using token cross-entropy loss ($\mathcal{L}_{\text{CE}}$) 13.
- **In SatQuery AI:** Used in the **Grounding Specialist** so users can type "highlight the water body" and the AI outputs precise coordinate boxes around the requested feature 13, 14.

Difference-Attention End-to-End Change Loss ($\mathcal{L}_{\text{change}}$)

- **How it works:** Used for comparing two photos taken at different times (T_1 vs. T_2). A **Siamese Neural Network** processes both photos side-by-side using "Difference-Attention" to focus on real structural changes (like newly constructed

roads or flood damage) while ignoring minor shadows 15, 16. The loss measures how accurately the AI identifies and describes those differences 16.

InfoNCE Contrastive Alignment Loss ($\mathcal{L}_{\text{InfoNCE}}$)

- **How it works:** Teaches the AI that an optical photo of a forest and a Synthetic Aperture Radar (SAR) scan of the exact same forest represent the **same real-world location**, even though they look visually distinct 3, 9. The loss pulls matching optical and radar feature vectors together while pushing non-matching regions apart 3, 9.
- **In SatQuery AI:** Powers the **Optical-SAR Fusion Pipeline**, enabling the system to cross-reference radar and optical data (e.g., identifying a river through thick clouds using radar) 9, 17.

4. Summary of Key Technical Concepts

Concept, What it does in simple terms

LoRA (Low-Rank Adaptation), "Attaches lightweight "plug-in" modules ($<1\%$ parameter size) to a single frozen AI model, creating specialized tools for VQA, Grounding, and Change Analysis without re-training the whole network 18, 19."

mIoU (Mean Intersection over Union), "A metric (0% to 100%) measuring how accurately a predicted shape or region mask overlaps with the real ground-truth boundary 3, 9."

BLEU / CIDEr, Automated grading metrics that compare AI-generated text descriptions against expert human descriptions to score naturalness and accuracy 20.

ECE (Expected Calibration Error), "Measures how honest the AI's confidence score is. If the model claims 90% confidence, it should actually be correct 90% of the time 3, 21."

💡 *Would you like to walk through a concrete example of how a user query travels step-by-step through the Query Interpreter and Compatibility Validator?*